Lab 06

Common Emitter Transistor Amplifier

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6.1 Objective

To investigate the performance of single-stage common emitter BJT amplifier

6.2 Introduction to BJT Amplifiers

One of the most useful applications of a transistor is amplification. With an amplifier, the output sinusoidal signal can be achieved greater than the applied input sinusoidal signal. When BJT is operated in active mode, changes in base-emitter voltage give rise to change in collector current. Voltage amplification is obtained utilizing this effect.

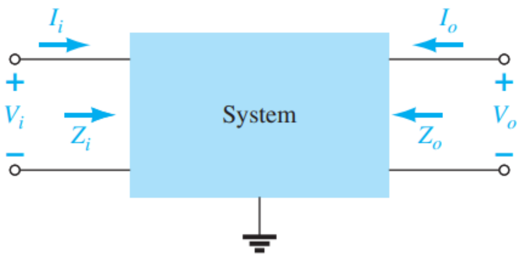


Figure 1: Two-port network representation of amplifier showing its parameters

Biasing (application of DC voltage) is required to make BJT operate in active mode at desired operating point before AC signals are applied for amplification. The superposition theorem is applicable for the analysis, permitting the separation of the analysis of the DC and AC responses of the system. In other words, one can make a complete DC analysis of a system before considering the AC response. Once the DC analysis is complete, the AC response can be determined using a complete AC analysis.

6.3. Pre-Lab Task

Task 1:

1. **What is the significance of small signal analysis in transistor amplifiers?**

Linear Approximation: Transistors operate non-linearly in general, but small signal analysis simplifies their behavior by treating them as linear components around a specific, desired operating point (Q-point). This makes calculations easier.
Frequency Response: It helps in understanding how an amplifier responds to different frequencies, like the cut-off frequency where the gain starts to decrease.
Gain Calculation: It helps determine the voltage gain, current gain, and power gain of the amplifier.
AC Equivalent Circuit: Small signal analysis involves replacing the transistor with its AC equivalent circuit, which consists of small-signal parameters like representing the transistor's behavior under small signal conditions.

2. **Why are Coupling and Bypass Capacitors used? Discuss their behavior during AC and DC input signals.**

Coupling Capacitors:

-Purpose: These capacitors are used to block DC components while allowing AC signals to pass from one stage of an amplifier to another**.**

-Behavior:

-DC Input: The capacitor acts as an open circuit, preventing any DC biasing voltage from transferring between stages.

-**AC Input:** It behaves as a short circuit (low impedance), allowing AC signals to pass. This enables signal transmission while isolating different DC biasing levels.

Bypass Capacitors:

-**Purpose:** Bypass capacitors are used to reduce noise and voltage spikes.It provides a low-impedance supply, thereby minimizing the noise. It bypasses noise to ground.

- Behavior:

- AC Signals: The capacitor effectively shorts AC noise or unwanted fluctuations to ground, preventing them from interfering with circuit operation.

- DC Signals: The capacitor acts as an open circuit, not affecting the steady DC voltage.

3. **List down the advantages and applications of CE Amplifiers.**

Advantages of CE Amplifier:

1. **High Voltage Gain** – Provides significant amplification of input signals.
2. **Moderate Current Gain** – Offers a good balance between voltage and current amplification.
3. **High Power Gain** – Due to high voltage and current gain, it is widely used in amplification stages.
4. **Good Input-Output Impedance Matching** – Moderate input impedance and low output impedance make it useful in various circuit designs.
5. **Phase Reversal** – The output signal is 180° out of phase with the input, which is useful in certain applications.
6. **Wide Frequency Response** – Can amplify a broad range of frequencies, making it suitable for audio and RF applications.

Applications of CE Amplifier

1. **Audio Amplifiers** – Used in sound systems and microphones to amplify weak signals.
2. **Radio Frequency (RF) Circuits** – Common in radio receivers and transmitters.
3. **Sensor Signal Processing** – Amplifies weak sensor signals for further processing.
4. **Oscillator Circuits** – Used in waveform generation circuits.
5. **Communication Systems** – Found in transmitters and receivers for signal amplification.
6. **General-Purpose Amplifiers** – Used in many low-power electronic circuits.
7. **Signal Processing** – Helps in processing and amplifying weak signals in different electronic applications.

6.4. Common Emitter Transistor Amplifier

Common emitter amplifier configuration is widely used. It provides large voltage gain (typically ten to hundreds) and moderate input and output impedance. CE amplifier circuit is shown in Figure 2.

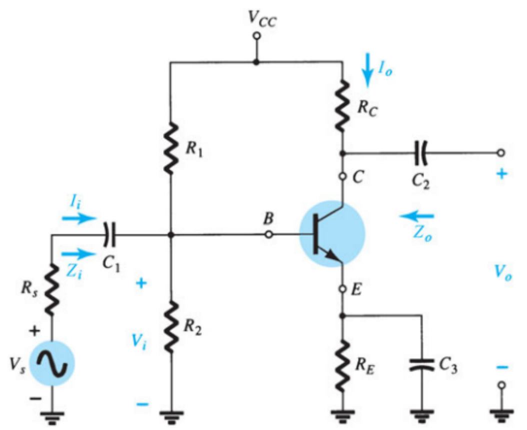


Figure 2: Common-emitter BJT amplifier circuit

Using a DC source and biasing configuration, the BJT is first biased to establish operating point in active region. Then, AC signal *Vs* is applied at input port through a capacitor. The source resistance is *Rs* here which will be neglected in our analysis for now. So, applied input to amplifier *Vi* is source voltage. The output voltage *Vo* is measured from collector terminal. The capacitors C1 and C2 are called coupling capacitors. The emitter resistor is bypassed through C3 bypass capacitor for AC signal.

For a particular DC operating point, when AC input signal *vi* is superimposed on DC voltage V_{BE}, the corresponding instantaneous curve of base-emitter voltage is shown with instantaneous base current. The corresponding instantaneous collector-emitter voltage is represented in i_c-v_{ce} curve which shows large variation in collector voltage by small changes in applied input AC voltage. The amplification is obtained at output node i.e. collector. From this, you can see the importance of locating Q point of BJT. If V_{CE} is very small, the BJT will enter saturation region as output voltage swings across it and if closer to Vcc, it will enter cut-off region. The location of bias point should allow swing in both directions equally, otherwise, output waveform peaks (positive or negative) will be clipped.

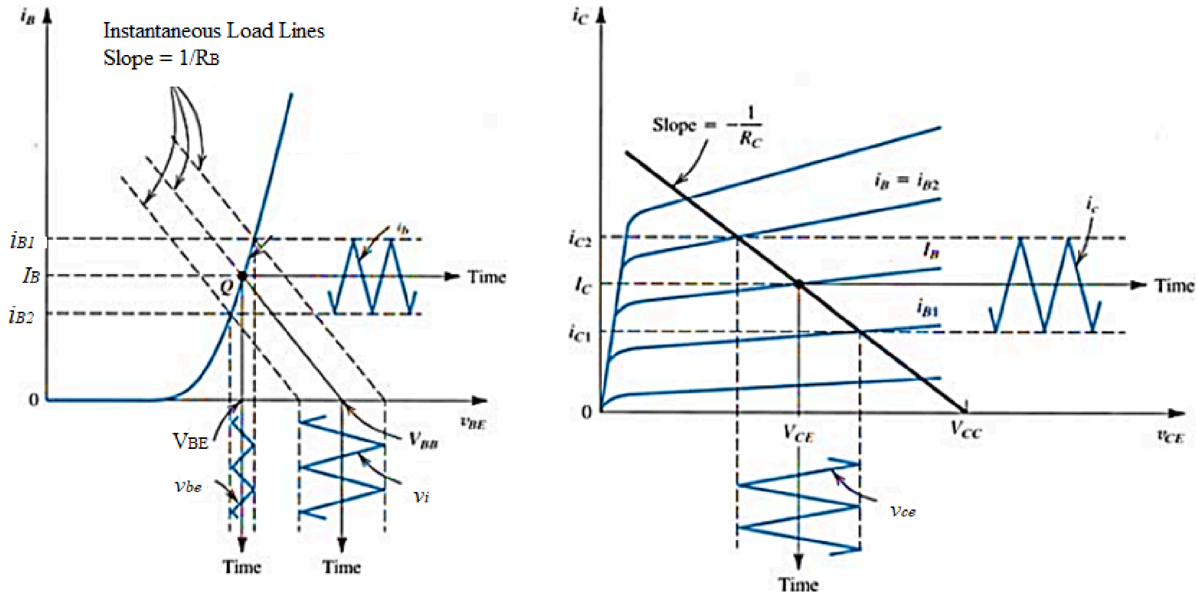


Figure 3: Graphical representation of AC signal components superimposed on DC

DC analysis has been covered in previous labs. For AC analysis, equivalent models are used. Here, for CE configuration r_e **model** will be used as shown in Figure 4.

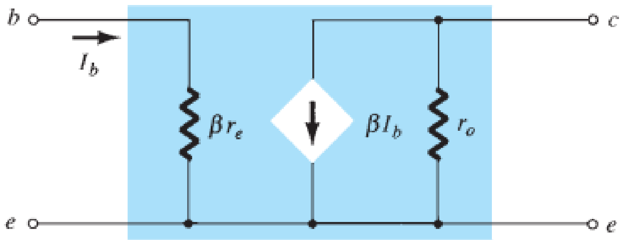


Figure 4: BJT AC equivalent model for CE configuration

Moreover, the AC equivalent of complete transistor network is obtained by:

- Setting all dc sources to zero and replacing them by a short-circuit equivalent
- Replacing all capacitors by a short-circuit equivalent
- Removing all elements bypassed by the short-circuit equivalents introduced by steps 1 and 2

Using the equivalent model of BJT, the AC equivalent of the network shown in Figure 2 is drawn in Figure 5.

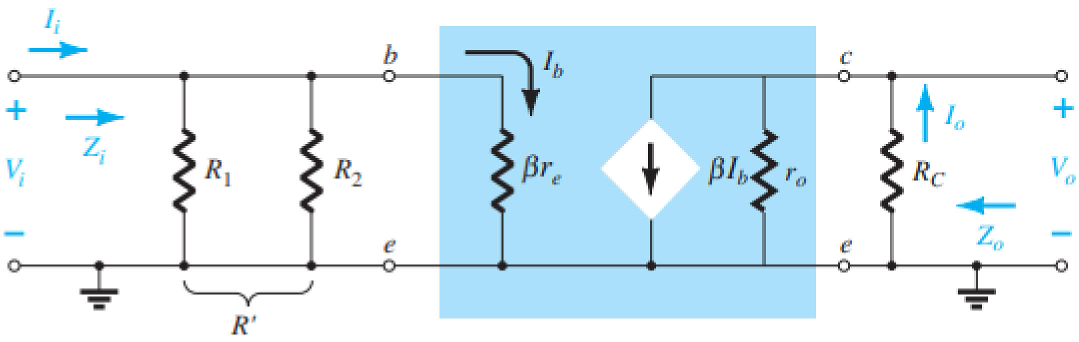


Figure 5: AC equivalent of CE amplifier network shown in Figure 2

6.5. Characteristic Parameters of Common Emitter Amplifier

The characteristic parameters of an amplifier include its voltage gain *Av*, input impedance *Zi*, output impedance *Zo* and current gain *Ai*.

Transistor AC dynamic resistance r_e shown in model can be calculated as $r_e = \frac{V_T}{I_E} = \frac{26mV}{I_E(mA)}$

- AC Voltage Gain *Av* is defined as $A_v = \frac{V_o}{V_i}$

  Here, *Vo* and *Vi* are output and input voltage which can be measured as rms, peak or peak-to-peak. From model, the AC voltage gain of a CE amplifier (under no-load) can be calculated using:

$$A_v = \frac{-R_C}{R_E + r_e}$$

  With bypass capacitor, use R_E =0 in the above equation which results in

$$A_v = \frac{-R_C}{r_e}$$

- AC Input Impedance, *Zi,* is that of the amplifier as seen by the input signal. For CE amplifier, it can be given as

$$Z_i = R_1 \parallel R_2 \parallel \beta(R_E + r_e) = R_1 \parallel R_2 \parallel \beta(r_e)$$

- AC Output Impedance, *Zo,* for CE amplifier can be approximated to

$$Z_o = R_C$$

Task 2: To perform DC analysis for CE BJT Amplifier

1. For the common-emitter amplifier circuit shown in Figure using BJT model 2N3904, perform DC analysis to complete Table 1. The supply voltage V_{CC} is 20 V. Before, performing DC analysis identify the BJT bias configuration and verify the conditions that should be satisfied for of the chosen analysis. Use DC Gain β equal to 300 in your calculations if needed.
   The circuit component values are given here:
   R_C =3kΩ, R_E=1kΩ, R_1=33kΩ, R_2=10kΩ, C_1= C_2=15μF and C_3=100μF.

Table 1: DC analysis of Common-Emitter BJT amplifier

Resistances	Measured*	DC Parameters	Calculated	Measured
R_C	2.81kohms	V_B	4.65	4.47
		V_E	3.95	3.8
R_E	0.94kohms	I_E	3.95mA	3.46mA
		V_C	8.15	9.0
R_1	32.18kohms	I_C	3.95mA	3.78mA
		I_B	13.2*10^-6	11.5uA
R_2	9.87kohms	**DC Gain β**	***300*** (Given)	328.69

2. Now, implement the circuit and verify the DC analysis by measuring the values for all the DC parameters listed in Table 1.

Task 3: To perform AC analysis for Common Emitter BJT Amplifier

For the same circuit, now perform AC analysis and calculate the parameters listed in Table 2. Use measured values in your calculations for AC analysis and fill in second column.

Table 2: AC analysis of Common-Emitter BJT amplifier

Characteristic Parameters	Calculated Values	Measured Results
Transistor AC Dynamic Resistance: r_e	6.58 ohms	6.5 ohms
Input Impedance: Z_i	1.57kohms	1.6khms
Output Impedance: Z_O	3kohms	2.93kohms
AC Voltage Gain: A_V	-456	-398.32

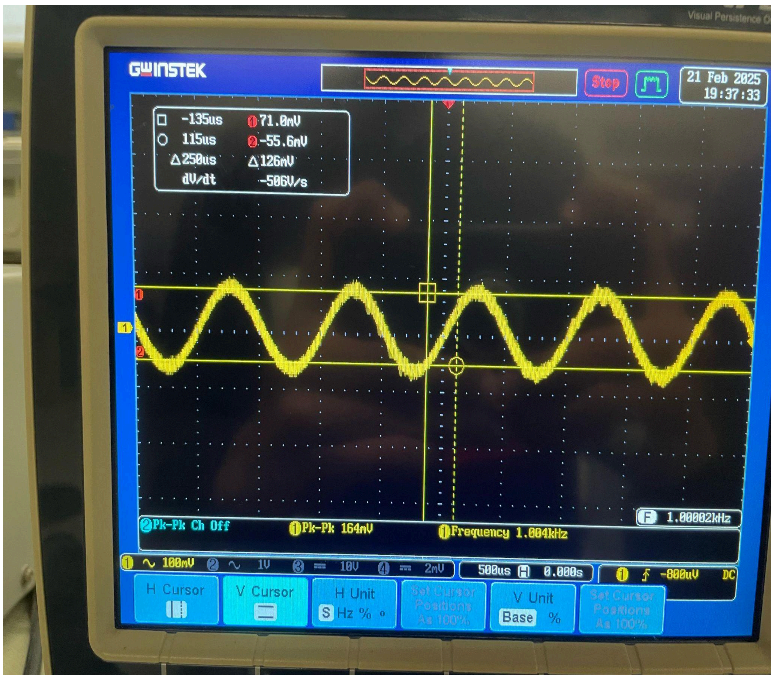
Task 4: To implement Common Emitter Amplifier Circuit

1. In the implemented circuit, now apply AC signal of magnitude 50mV pk sine wave at 1k Hz through coupling capacitor of 15µF.
2. Use bypass capacitor of 100µF at emitter and measure the output voltage through another coupling capacitor of 15µF as shown in Figure 2. This is the unloaded output of amplifier as there isn't any load resistor connected at the output terminal of amplifier. Write down your observations in Table 3.
3. Observe the output voltage waveform. Analyze the magnitude and phase of this amplifier output voltage with respect to the applied AC input signal.

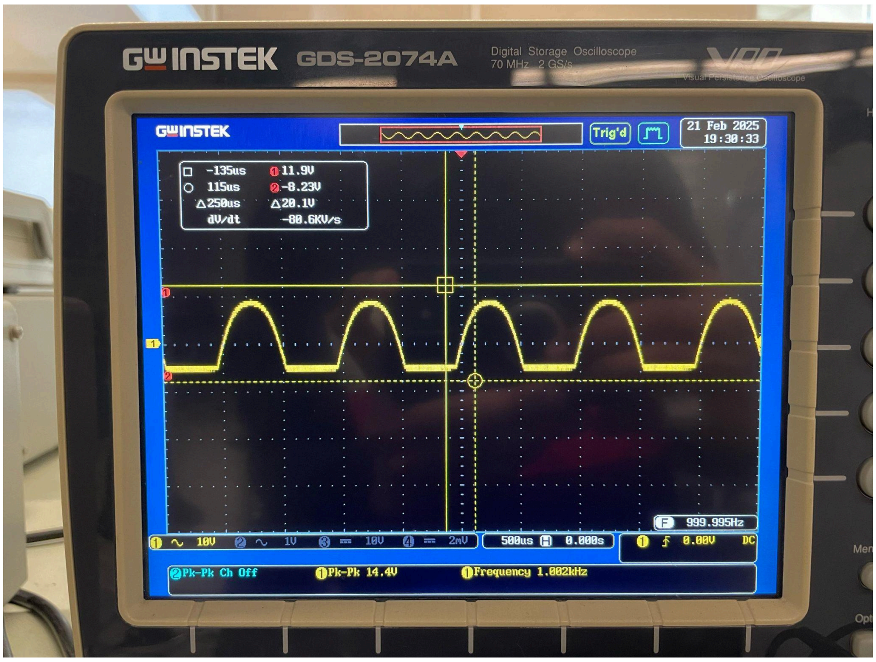
Magnitude:
The magnitude is amplified as compared to the input signal. The peak to peak voltage of the input waveform is in millivolts, whereas the output is around 14V.
Phase Shift:

The output waveform is phase shifted between 175-180 degrees and ideally, it would be 180 degrees.

Input Waveform:



Across C2 (output waveform):



Task 5: To experimentally determine the characteristic parameters of CE amplifier

The CE amplifier characteristic parameters discussed in Section 6.5 will now be experimentally determined and compared with the ones calculated in Task 3.

● Amplifier Voltage Gain: *Av*

1. Using the measured values of input and output voltages of the amplifier, determine the magnitude of amplifier voltage gain and note it down in Table 2.
2. Keeping the phase difference in mind, assign its sign (+/-).

● Amplifier Input Impedance: *Zi*

1. To determine the input impedance of this amplifier, connect an input measurement resistor, *Rx*=1kΩ, as shown in Figure 6. Apply the same input signal (50mVpk at 1 kHz) but this time through this known resistance. Measure *Vi* i.e., the signal after this resistor *Rx*. Note down in Table 3.

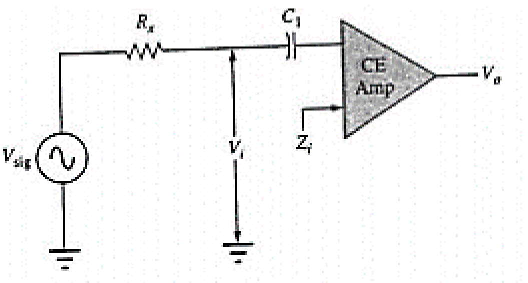


Figure 6: Circuit for determining input impedance

2. Mathematically, for this circuit this voltage *Vi* can now be expressed as

$$V_i = \frac{V_{sig}}{Z_i + R_x} Z_i$$

   Solve the given equation for *Zi*.

Vsig Zi= Vi(Zi+Rx)

Vsig Zi = (ViZi + ViRx)

Vsig Zi- ViZi = ViRx

Zi(Vsig - Vi) = ViRx

Zi = ViRx / (Vsig-Vi)

3. Determine the value of input impedance. Note down in Table 2.

Table 3: Data table for experimental determination of CE amplifier characteristic parameters

Characteristic Parameter	Required Measurements	Measured Values
For CE Amplifier Voltage Gain	Input Voltage: *Vi*	50mV
	Output Voltage (Unloaded): *Vo*	10.8V
For CE Amplifier Input Impedance	Measurement Resistor: *Rx*	0.995kohms
	Source Voltage: V_{Sig}	51mV
	Input Voltage: *Vi*	32mV
For CE Amplifier Output Impedance	Output Voltage Unloaded: *Vo*	11V
	Load Resistor: R_L	2.93k
	Output Voltage Load: V_L	5.4V

- ## Amplifier Output Impedance: *Zo*

1. Remove the input measurement resistor *Rx*.
2. Apply AC input signal (50mV pk at 1 kHz) directly. Measure the unloaded output voltage *Vo*.
3. Now, connect load resistor of 3kΩ (use measured resistance) and measure the corresponding loaded output voltage. Note it down in Table 3.
4. When loaded, the output voltage can be expressed as

$$V_L = \frac{R_L}{Z_o + R_L} V_o$$

Solve the above for output impedance *Zo*.

$$Zo = \frac{R_L V_o}{V_L} - R_L$$

5. Calculate the output impedance to complete Table 2.

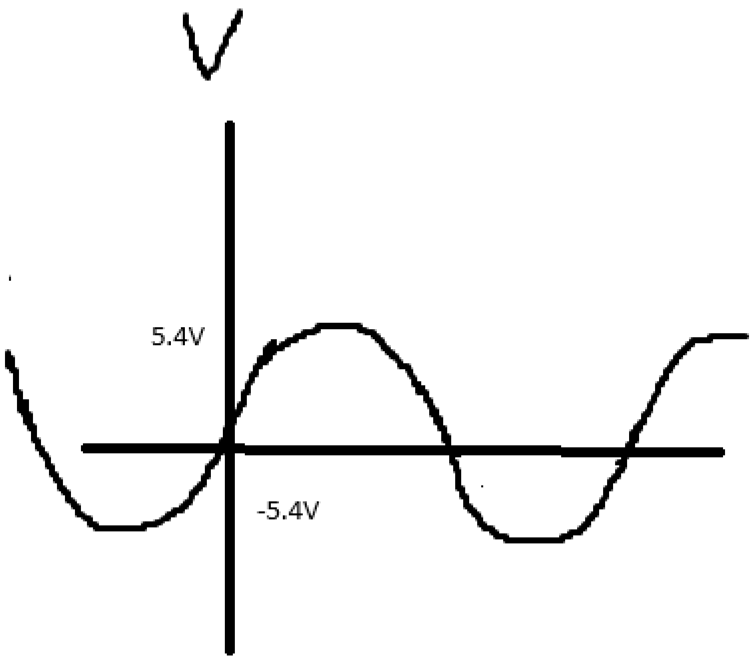
Task 6: To investigate the performance of CE amplifier

1. Using the data collection of Table 2, compare the estimated and measured characteristic parameters. What are your comments about the approximations considered in the analysis?

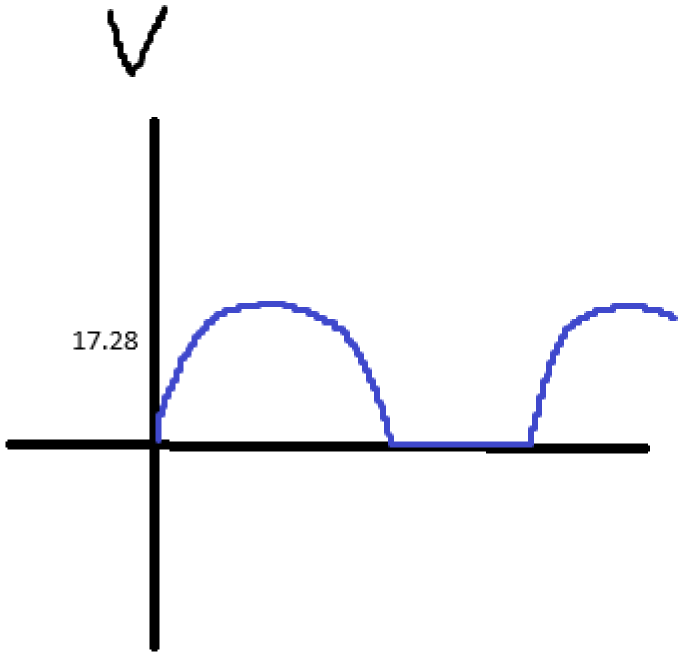
The measured parameters are similar to the estimated values, though minor deviations remain due to resistor tolerances, transistor β variations, and other technical factors. The negative sign in the voltage gain confirms our analysis that the output is inverted.

2. Now, in the actual CE amplifier circuit where there is no output load or input measurement resistance connected in the circuit having the applied input of 50mVpk sinewave at 1k Hz, obtain the output voltage waveform and observe this output as you gradually increase the input signal voltage from 50mV to 200mV. Analyze the changes you observe in amplifier output voltage waveform.
   With a 50mV input, the output is a clean, amplified sine wave. As the input increases, the output amplitude rises proportionally. However, when the input approaches 150mV, clipping begins at the peaks, indicating that the transistor has reached saturation and cutoff. Beyond this, up to a 200mV input, the waveform becomes distorted.

At 50mV:



At 200mV:



3. Try measuring the output before the coupling capacitor connection at output node. Justify the need of coupling capacitor?
   A coupling capacitor is necessary to block the DC bias, allowing only the AC signal to pass through. This prevents damage to the next amplification stage and ensures proper operation in multi-stage amplifier circuits.
4. If you increase the resistor at collector terminal to 5kΩ, what changes do you observe in the output voltage waveform? Justify your analysis. *Hint: Observe the DC operating point Vc too to relate.*
   Increasing Rc to 5kΩ results in a greater voltage drop, leading to a lower Vc. This may cause the transistor to enter saturation, leading to distortion. This analysis is supported by the equation Vc = Vcc – IcRc, which illustrates the impact of increasing Rc.

6.6. Post Lab Task

Task 7: To calculate current of the CE amplifier

1. Find the current gain of the amplifier implemented in lab when it is loaded.

   A = Ic/Ib = 3.78mA/11.5uA = 328.7

2. From voltage gain and current gain, one can say that the output AC power is greater than the input AC power. How can you justify this when conservation of energy dictates that over time the total power output, Po, of a system cannot be greater than its power input, Pi, and that the efficiency defined by efficiency cannot be greater than 1?

   The output AC power appears greater than the input AC power because the amplifier draws additional energy from the DC power supply, converting it into an amplified AC

signal rather than creating energy. This process does not violate the conservation of energy, as the total power input includes both the AC signal and the DC supply. However, due to losses in resistors and other components, the overall efficiency remains below 100%.

Assessment Rubric

Lab 06
Common Emitter Transistor Amplifier

Name:	**Student ID:**

Points Distribution

Task No.	**LR1 / LR2** Simulation / Circuit 36	**LR 4** Data Collection 16	**LR 6** Calculation 24	**LR 10** Analysis 20
Task 1	-	-	-	/4
Task 2	/8	/4	/8	-
Task 3	-	-	/8	-
Task 4	/8	/4	-	-
Task 5	/12	/4	/4	-
Task 6	/8	/4	-	/8
Task 7	-	-	/4	/8
CLO Mapped	CLO 1	CLO 1	CLO 1	CLO 1
Total				

Affective Domain Rubric		**Points**	**CLO Mapped**
AR 4	Report Submission	/4	CLO 4

CLO - LDL	**Total Points**	**Points Obtained**
CLO 1 – (Psy-3)	96	
CLO 4 – (Aff-3)	4	
Total	**100**	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.